

DESIGN DOSSIER

Heli-Town Hell

GAME DESIGN & FUNCTIONAL BLUEPRINT

Detailed project review and phased Godot development plan

A single-player, 3D arcade helicopter combat game about flying through a hostile city, surviving escalating battles, completing tactical tasks, collecting upgrades, and deciding when to extract with earned rewards.



Brief overview

Play: steer the helicopter separately from its chin gun; fight ground and air units; manage hull, fuel, ammunition and flares. **Progress:** collect run XP, improve weapons, complete missions and deliveries, then invest credits in the hangar. **Finish:** extract, defeat the heavy gunship and leave safely, or continue into Endless Overdrive.

Development target: Godot 4.7.2 stable, verified against the official download page on 28 September 2026. The roadmap is a proposed rebuild plan, not an existing Godot implementation. [G1]

Prepared from the supplied project. Existing implementation languages, libraries, hosting and tooling are intentionally omitted. Game rules, player-facing functions and Godot development activities are included.

Evidence: Review date: 28 September 2026 | Existing project screenshot

[START HERE](#)

Reading guide & contents

Read the overview first, then use this index to explore a system or development phase. Current gameplay, implementation gaps and proposed rebuild work are distinguished throughout.

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How to interpret the document

Current: behavior traced to active source paths. **Defined / intended:** a setting or description exists, but integration may be incomplete. **Proposed:** Godot work or acceptance requirements.

Distances use world units; the current interface often labels them meters. Existing images are references. The roadmap targets the latest stable release verified on the review date.

Evidence: Source evidence and interpretation are distinguished throughout

GAME DESIGN

The player experience & run loop

The core promise

The player pilots a gunship through a populated warzone while choosing between immediate survival and optional reward opportunities. Flying, aiming and target selection are separate skills. Buildings provide cover and constrain routes; ground defenses control space; aerial enemies create attack windows.

One complete run

- **Prepare:** choose an aircraft, difficulty, settings, permanent upgrades and one modification per weapon. New profiles receive a one-time 150-credit grant.
- **Learn and launch:** the first-run tutorial introduces flight, aiming, fire, altitude, evasion, flares, salvo, super and pause. A protected opening leads into timed waves.
- **Fight and grow:** defeat enemies, collect XP gems and useful drops, reload, refill fuel and select one of three run upgrades when leveling up.
- **Choose diversions:** destroy installations, escort or defend a location, rescue a crew, hunt elites, or carry cargo between depots.
- **Make the exit decision:** from wave 6 and after two destroyed objectives, a once-per-run extraction opportunity can appear. Wave 10 introduces the heavy gunship; defeating it offers safe extraction or Endless Overdrive.
- **Debrief:** inspect score, kills, accuracy, survival time, objectives, rewards and losses. Purchase improvements and start again.

Design pillars

Readable pressure: many enemies may exist, but attack permissions limit simultaneous fire. **Movement mastery:** heading, momentum, altitude and turret aim matter. **Risk and reward:** unsecured bonuses and salvage create an extraction decision. **Short-term variety:** missions, districts, wave themes, modifiers and upgrade choices change the run.

DESIGN NOTE

This is a run-based combat game. A story campaign, multiplayer, online leaderboard and character dialogue system were not found in the reviewed gameplay. Do not assume these are part of the rebuild scope.

Evidence: engine.ts; App.tsx; mission.ts; delivery.ts; logic.ts

GAME DESIGN

Controls & supported input

Action	Current desktop control
Turn the helicopter	A / D or Left / Right arrows
Forward / reverse thrust	W / S or Up / Down arrows; relative to the nose
Strafe sideways	Q / E; relative to the helicopter heading
Climb / descend	Shift or Space climbs; Ctrl descends
Aim / primary fire	Move mouse to aim chin turret; hold left mouse to fire selected weapon
Paint / release salvo	Hold right mouse to accumulate locks; release to launch
Flares	C; requires a charge and a completed cooldown
Weapon / reload	1-4 or mouse wheel; R reloads
Pause / restart	Escape or P pauses; Enter can restart from results

Controller behavior

Left stick supplies yaw and thrust. D-pad left/right strafes; up/down changes altitude. Right stick aims independently and also fires when deflected. A, right trigger or right shoulder can fire; X reloads; left shoulder deploys flares. The active gameplay mapping reviewed does not provide parity for every desktop action.

Touch behavior

Two virtual sticks cover flight and aiming, with visible Fire, Flares and Super buttons. The left stick supplies heading-relative yaw/thrust. The current touch overlay does not expose a complete set of altitude, strafe, salvo and weapon controls; treat mobile input parity as unfinished.

DESIGN NOTE

Retired mappings must not be taught: Q is no longer salvo, Shift is no longer afterburner, and double-tap dash / manual camera recenter bindings have been removed. Devastation remains available through the touch Super button, but has no current desktop keyboard binding.

Evidence: engine.ts: input handlers and pollGamepad; App.tsx: help and touch controls

GAME DESIGN

Flight, aircraft & camera

Authoritative flight behavior

Current flight uses a heading-relative controller. Turning changes the nose direction; forward/reverse thrust follows that direction; strafing follows the local right/left direction. Combined thrust and strafe are normalized, so diagonals do not grant extra input strength. Releasing thrust leaves momentum that decays into a hover.

Baseline property	Current value
Horizontal cruise cap	72 world units/second
Thrust acceleration / drag	420 units/second ² ; exponential drag coefficient 1.35
Maximum yaw rate	1.85 radians/second, approximately 106 degrees/second
Climb / sink / vertical acceleration	6.5 / 7 / 9 units per second or per second ²
Starting altitude / ceiling	22 units / absolute 60-unit ceiling
Surface clearance	2.4 units above reachable terrain or roof
Visual pitch / bank limits	Approximately 18 / 28 degrees

Obstacles, damage and cargo

Intact buildings block flight. Reachable roofs raise the safe floor; towers above the flight ceiling must be avoided. Damage to the engine reduces thrust and vertical demand, while rotor damage reduces yaw response. Cargo affects acceleration and lift; it does not reduce the current hard 72-unit horizontal cap. Explosion impulses add displacement.

Camera and aircraft selection

The camera follows behind the heading, pulls back with speed, looks ahead of the nose and widens its field of view slightly. Buildings between the camera and aircraft fade visually while retaining collision. Apache, Nighthawk and Warlock have different silhouettes; the older model-specific handling multipliers live in a disabled controller branch, so distinct current flight performance is not established.

DESIGN NOTE

Arcade/Simulation, speed boosts, engine/rotor upgrade promises and wind influence need a deliberate compatibility pass. Several effects are represented in settings or older logic but are not fully applied by the current flight controller.

Evidence: entities.ts: FLIGHT_SPEC and applyHeadingRelativeFlight; engine.ts: updateCamera

GAME DESIGN

Weapons & firing functions

Weapon	Damage per projectile	Interval	Ammo / reload	Payload
Machine Gun	13	0.055 s	300 / 1.5 s	2 rounds; speed 430
Missile	55	0.95 s	20 / 2.5 s	1 projectile; speed 260; blast radius 16
Rocket	80	1.45 s	12 / 3.2 s	1 projectile; speed 235; blast radius 28
Shotgun	10	0.45 s	40 / 2 s	6 pellets; speed 280; spread 0.3

These are factory values before mastery, modifications, temporary bonuses and target affinity. An interval is the time between trigger events. One ammunition unit is consumed per trigger event, including multi-projectile shots.

Weapon roles

Machine Gun: sustained precision and aerial tracking. Rounds leave the physical chin-gun muzzle along the barrel direction. **Missile:** homing pressure against armored targets and boss hulls. **Rocket:** larger explosive attacks against armor and structures, with a recoil shove. **Shotgun:** multiple close-range hits and light-target control.

Firing and reload sequence

The selected weapon must exist, have ammunition, be outside its fire interval and not be reloading. Firing consumes one round, records a trigger event, launches its projectiles and plays feedback. Empty ammunition automatically starts a reload; R starts it manually. Switching weapon cancels the current reload. Reload replenishes the magazine rather than spending a separate finite reserve inventory.

Aiming and collision

Auto-aim chooses valid nearby targets and tracks the gun, independently of the helicopter heading. Bullets do not home. Non-machine-gun paths can add projectile assistance; despite the Rocket being described as unguided, its current firing path can supply steering when a lock target is found. Explosions can damage nearby enemies and buildings.

DESIGN NOTE

For the rebuild, decide whether rockets should remain assisted or become strictly ballistic, and communicate that choice consistently in the armory and tutorial.

Evidence: types.ts: WEAPON_CONFIGS; engine.ts: fireWeapons, reload and hit resolution

GAME DESIGN

Salvo, mastery & weapon mods

Lock-on salvo

Holding right mouse paints up to six total locks, with up to three on the same enemy. The search radius around the aim point is 38 units and another lock can be painted every 0.18 seconds. Releasing fires one homing missile per lock: 55 damage, speed 265, blast radius 12. This uses a separate 5-second cooldown rather than the selected weapon magazine. Salvo Overclock reduces it, with a 1.5-second floor.

Weapon experience

Kills award one weapon XP to the last-fired weapon. Weapon levels run from 1 to 5; actual cumulative thresholds are **0, 6, 14, 24, 36 XP**. Each level above 1 adds 18% damage, reduces the firing interval by 9% of baseline and reduces reload time by 12% of baseline. Level 4 adds one projectile. Level 5 adds signature effects and +10 maximum hull with a repair.

At level 5 the machine gun gains another 25% damage; missile gains another projectile beyond its level-4 bonus; rocket blast radius becomes at least 40; shotgun gains another 15% damage. The highest mastery level is recorded for the hangar, while each new run resets weapon XP and active levels.

Weapon	Mod A	Mod B
Machine Gun	Incendiary: 18% burn chance	Piercing: +35% damage vs tank and boss hulls
Missile	EMP Warheads: weapon disruption	Cluster Warheads: three mini-blasts
Rocket	Napalm: +30% blast radius and burning	Shaped Charge: +45% vs bosses / elites
Shotgun	Shock Shells: 25% slow chance	Slug Barrel: +30% damage; spread x0.55

Choose factory configuration or one of two mods independently for each weapon. Choices persist; the selection path does not charge a purchase price. Mod effects combine with run bonuses, subject to applicable status caps.

DESIGN NOTE

Mastery XP is attributed to the last-fired weapon, not reliably to the projectile that caused the kill. Fix attribution in Godot so delayed missiles, burns and environmental kills cannot level the wrong weapon.

Evidence: logic.ts: weapon XP, level bonus and WEAPON_MODS; engine.ts: grantWeaponXp and fireWeapons

GAME DESIGN

Damage, defenses & status effects

Hull and survivability

Base hull is 100. Permanent armor adds 10 maximum hull per rank; permanent and run armor contribute damage reduction, with the final resolver capping reduction at 24%. The Rotor Armor perk also contributes to this same capped resolution. Shields, active dash logic, opening protection and Devastation can block damage. Damage sources include bullets, explosives, missiles, collisions and fuel starvation.

Weapon affinity	Light	Armored	Structure	Air	Boss core
Machine Gun	1.10	0.85	0.80	1.25	1.00
Missile	1.00	1.45	1.35	1.00	1.30
Rocket	1.20	1.40	1.40	0.90	1.00
Shotgun	1.30	0.90	1.00	1.15	1.00

Values are damage multipliers in the affinity rules, before other bonuses. The benefit depends on the target category assigned by the hit path. A broad weapon label is not a guarantee that every projectile path uses identical targeting or status behavior.

Elemental effects

Burn: 14 damage/second for 3 seconds. **EMP:** disables enemy attacks for 2 seconds. **Shock:** multiplies movement speed by 0.55 for 2.5 seconds. Mods and upgrade stacks can apply these effects; combined proc chance is capped at 95%. The EMP Pulse pickup applies the shared EMP status, so its nearby enemy disruption is 2 seconds despite an older 4-second comment.

Fuel pressure

The base tank holds 100 fuel and drains 0.85/second during live play. Permanent fuel upgrades add capacity and reduce drain; fuel run upgrades, Lean Burn and engine ranks further improve endurance. At zero fuel, the aircraft takes repeated starvation damage rather than instantly ending the run. Fuel pickups and depots allow recovery.

Subsystem damage and repair

A helicopter damage call has a 40% chance to damage either the engine or rotor by half the damage amount. Both start at 100. A fully damaged engine can halve its efficiency; a fully damaged rotor can reduce its efficiency to 40%. Repair also restores subsystem condition.

Casual difficulty grants one emergency repair at critical hull. Opening protection preserves the initial full tank and hull. These are distinct from repair pickups.

Evidence: combat.ts; logic.ts: calculateDamageAffinity and statuses; engine.ts: player damage and fuel

GAME DESIGN

Enemy roster & combat roles

Five base families support the roster: infantry clusters, shooter aircraft, tanks, light aerial units and the heavy gunship boss. Base hull values are 25, 30, 100, 22 and 260 respectively, before variants, elites, waves, themes and difficulty scaling.

Variant	Ordinary gate / cap	Role and player response
Scout Drone	Wave 2 / 10	Fast light aircraft. Track with gunfire; watch its approach.
Flak Tank	Wave 3 / 6	Ground anti-air pressure. Break line of sight and use explosives.
Missile Carrier	Wave 5 / 2	Armored missile threat. Save flares and remove it early.
Attack Gunship	Wave 6 / 8	Durable attack-run aircraft. Punish committed passes.
Shield Drone	Wave 6 / 2	Protects allies. Prioritize the support unit.
Rocket Gunship	Wave 7 / 5	Heavier ranged attack. Read its launch setup.
Repair Drone	Wave 7 / 1	Repairs nearby allies. Kill before extended damage trades.
Heavy Gunship variant	Wave 7 / 1, rare	Tough elite-sized aircraft; separate from the boss.
Kamikaze Drone	Wave 8 / 6	High-speed collision attack. Evade the committed line.
Siege Tank	Wave 8 / 1, rare	Slow heavy armor and high damage. Flank through cover.
Gatling Heavy	Wave 9 / 2	Tracking sustained-fire role on a tank base hull.
Interceptor	Wave 10 / 4	Fast strafe, peel-away and re-attack aircraft.
Minelayer	Wave 11 / 2	Slow platform deploying homing proximity mines.

Gates above belong to the variant definitions and ordinary selection. Special events and mission spawns can bypass them: an air-raid event can introduce kamikaze units earlier. Rare units are excluded from the ordinary random variant picker and require special selection paths.

DESIGN NOTE

Rebuild requirement: every enemy should have a recognizable silhouette, a readable attack tell, a useful counter and a visible reward. Resolve the Gatling Heavy aircraft-versus-tank description mismatch before art production.

Evidence: types.ts; logic.ts: ENEMY_VARIANTS; entities.ts; engine.ts: event spawning

GAME DESIGN

Enemy intelligence & modifiers

Air combat is a sequence

Entry → approach → setup → attack run → break away → reposition. Aircraft predict the player path, align before firing, commit to a pass and then leave the attack line. Their bodies follow flight direction while their gun pivots track separately. During an attack, limited turning lets a sudden player maneuver cause a miss. Terrain look-ahead adjusts altitude to avoid buildings.

Ground and support behavior

Tanks move to firing lanes, aim with a separate turret, fire, and reposition. Infantry remains grounded and fights as a cluster. Line-of-sight checks prevent intended shots through intact buildings. Shield drones protect allies; repair drones restore an ally at up to 5% of its maximum hull per second. Support roles need to be identifiable within a group.

Coordinated pressure

A combat director controls air attack slots, ground permissions, heavy attacks and directional sectors. Early attacks favor simpler directions; later waves add flanks and pincers. Waiting enemies continue moving rather than all firing at once. Short lulls allow repairs, pickups and repositioning. Deaths and timeouts release attack permissions.

Modifier	Gameplay consequence
Shielded	Extra protection absorbs damage before the hull; visible shield feedback.
Regenerating	Recovers hull when not recently damaged; maintain pressure.
Elite	Larger, tougher, higher-value enemy; non-boss base hull x1.7 and score x2.2 before other scaling.
Explosive	Available from wave 6: death detonation, radius 26, damage 30; nearby entities are at risk.
Splitter	Available from wave 8: death releases three drone escorts.
Vampiric	Available from wave 9: healing is linked to damage dealt to the player, nominally 50%.

DESIGN NOTE

A destroyed aircraft may enter a visible death spiral and crash on a roof or ground. Its XP and loot can be delayed until the crash; the kill must still be credited exactly once.

Evidence: combatDirector.ts; entities.ts: air/tank/support states; logic.ts: affixes; engine.ts

GAME DESIGN

Waves, difficulty & battlefield events

Waves advance on elapsed combat time, not by clearing every enemy. Duration is 45 seconds on wave 1, falling by 1.5 seconds per wave to a 30-second floor. Wave changes include a short transition and a modest repair of 10 + new wave number. Boss encounters have their own pacing and decision state.

Wave group	Baseline pressure and milestones
1-3	Population targets 5, 7, 10; early learning protection; wave 3 forces the storm theme.
4-6	Targets 12, 14, 16; radar reinforcement from wave 5; milestone elite pressure; heavy gunship variants enter ordinary roster from 6.
7-9	Targets 18, 21, 24; mixed arms, support aircraft and growing heavy pressure; guaranteed SAM presence checked from wave 8.
10	Heavy gunship boss; background population target drops to 10.
11+	Endless Overdrive after the boss choice; expanding roster and capped escalation; major bosses on later multiples of 10.

These targets are starting values, not guaranteed on-screen totals. Difficulty, threat, current intensity, themes and spawn constraints alter the result. Air attack slots grow from 1 early to 4 at wave 9; boss battles allow 2. Heavy simultaneous attacks rise from 1 to 3; boss battles restrict them to 1.

Difficulty	Hull / hostile damage	Population factor	Opening grace
Casual	0.75 / 0.45	0.65	2.5 s
Normal	1 / 1	1	2 s
Hard	1.25 / 1.25	1.12	1.5 s

Variation within the run

Swarm Tide favors more weak enemies; Armored Column favors tougher hulls; Frenzy accelerates arrivals; Storm Surge increases weather; Night Surge combines heavier pressure and atmosphere. Milestone waves omit themes. Events include contested salvage, missile storms, convoy ambushes and air raids. Night operations can roll from wave 8 at a seeded 25% chance.

Enemy scaling is intentionally bounded: normal hull growth reaches a 2.35x ceiling, damage 1.70x and speed 1.18x. Composition and simultaneous attack pressure should provide challenge without endless health inflation.

Evidence: logic.ts: difficulty and wave rules; engine.ts: startNextWave and triggerBattlefieldEvent

GAME DESIGN

Boss battle & continuation

The heavy gunship encounter

A warning and introduction precede the boss. The boss has its own heavy aircraft model, large hull, rotating gun assembly, engine damage presentation and reactor-core effects. A tank and infantry escort are queued on entry. The encounter uses the shared heavy-attack permission system so its most dangerous attacks do not overlap freely with other heavy weapons.

Player-facing stage	Hull band	Attack behavior
Opening stage	Above 66%	Internal phase 3; regular spread volley, up to five projectiles.
Escalation stage	Above 33% through 66%	Internal phase 2; stronger telegraphed volley of seven, plus radial-ring attacks.
Final stage	33% or less	Internal phase 1; stronger telegraphed volley of nine; faster, denser pressure.

Telegraphed attacks use a 0.8-second warning. Mid-stage volley speed is 115; final-stage speed is 135. The numbered internal phases count down, so tutorial and boss UI should use consistent chronological labels. Damage attachments communicate deterioration: engine damage, smoke, electrical arcs and an unstable core.

Defeat and reward

The defeat path awards the normal 320-credit boss bounty plus a separate 1,000-credit boss reward, 10 direct salvage and a 40-hull repair. XP and additional boss loot are also available. Threat bonuses remain unsecured where applicable. After a brief delay, the player chooses safe extraction or Endless Overdrive.

Endless Overdrive

Continuing starts wave 11 with a 1.25 multiplier. The mode increases combat pressure and score opportunity; additional salvage may drop. Later milestone bosses continue the cycle. Clearly distinguish this mode from the temporary damage pickup also called Overdrive.

DESIGN NOTE

Current Bomb/Airstrike logic destroys all active enemies, including the boss, with no boss exception. A Godot design decision is required: preserve this power fantasy, or replace boss instant death with large capped damage and show the exception in the item description.

Evidence: engine.ts: boss spawn, death, extraction and Overdrive; entities.ts: boss attacks; logic.ts

GAME DESIGN

Installations, SAMs & countermeasures

Target	Base hull / score	Destruction result
SAM site	170 / 300	55 secured credits; 5 salvage; threat +8; 18-second suppression of enemy fire; cache drop and nearby homing-threat diversion.
Radar tower	220 / 250	45 secured credits; 4 salvage; threat +7; 30 damage to active enemies; cache drop; removes its support presence.
Ammo depot	190 / 200	35 secured credits; 3 salvage; guaranteed Bomb pickup and selected-weapon refill.

SAM engagement sequence

Search → track → lock → fire → reload. Nominal detection range is 330 units; firing range is 38-255. The launcher must align before its lock progresses. Base lock time is 1.15 seconds; reload is 5.5 seconds. A missile travels at 155 units/second, turns toward its target and has a 4.8-second lifetime. Waves, difficulty and radar support modify engagement pressure.

Radar support can accelerate nearby SAM locks by 1.2x. Breaking the firing envelope or alignment cancels the lock. Lock-progress audio grows more urgent; the HUD and minimap expose SAM state and incoming missile danger. Rooftop gun turrets are separate ambient threats and disappear with their host building.

Flares

At rank 0: three charges, 6.5-second cooldown and a 1.05-second decoy window. Launching consumes one charge. Eligible homing missiles within 135 units are diverted; between 135 and 190 diversion becomes probabilistic; farther missiles are unaffected. Flares are timing-based defense, not universal invulnerability.

Countermeasure rank 3 adds a fourth charge; rank 2 lowers cooldown to 6 seconds; rank 4 extends the active window to 1.25 seconds; rank 5 lowers cooldown to 5.2 seconds and improves outer-range effectiveness. Countermeasure pickups, certain missions and completed ammunition deliveries replenish charges.

DESIGN NOTE

The radar destruction path applies damage but does not explicitly dispatch every resulting death through the shared reward routine at that point. Verify lethal radar-blast credit and loot in the rebuild.

Evidence: sam.ts; mechanics.ts; engine.ts: objective destruction, SAM launches and flares

GAME DESIGN

Tactical missions: all nine types

One tactical mission is active at a time, alongside the independent cargo contract. Generation starts after an initial delay, follows a nine-type sequence with availability fallbacks, and waits 20 seconds after completion. Main rewards are claimed once. Most main credit rewards scale with wave/threat up to 1.45x; the table shows unscaled bases.

Mission	Completion requirement	Base reward: CR / XP / salvage
Destroy SAM	Destroy the assigned battery.	300 / 18 / 2
Destroy Radar	Destroy the assigned radar.	350 / 20 / 2
Cargo Run	Complete the linked delivery contract.	0 / 12 / 2; delivery pays separately
High Value Target	Destroy the spawned, identified elite.	360 / 24 / 3; +1 flare
Clear Airspace	Kill 8-13 enemies within 140 units of the marked area.	240 / 20 / 2
Defend Beacon	Accumulate 25 seconds within a 34-unit radius.	320 / 22 / 3; +12 repair
Escort Convoy	Accumulate 26 seconds within 42 units of a convoy during its 42-second route.	340 / 24 / 3; +10 repair
Rescue Crew	Accumulate 18 seconds within 34 units of the crash-site marker.	300 / 20 / 2; +15 repair
Bounty Leader	Destroy the designated elite squadron leader.	420 / 28 / 5; +1 flare

Optional bonuses

SAM/Radar: avoid SAM missile damage for +100 credits and 1 salvage. Cargo: 10 kills while carrying for +80 and 1 salvage. HVT: finish within 65 seconds for +120 and 1 salvage. Bounty: 75 seconds for +150 and 2 salvage. Airspace/Defend/Escort/Rescue: finish above 50% hull for +75/+90/+95/+85 credits respectively, each plus 1 salvage.

DESIGN NOTE

A failed optional bonus does not fail the main mission. HVT/bounty deadlines currently govern bonus eligibility; they do not themselves prove a target escape. Defend/rescue are proximity timers, and escort is proximity to a scripted route rather than a fully simulated convoy-health objective.

Evidence: mission.ts; engine.ts: mission generation, elite spawning and reward claims

GAME DESIGN

Cargo delivery: full lifecycle

Route selection

Contracts connect distinct procedural depot hubs 320-1,580 units apart. Industrial and waterfront districts are the most likely depot locations; residential and forest districts are much less likely. Endpoints are deterministic for their world location, so unloading and returning to a region should not move the destination.

Step	Player action and rule
1. Offer	Follow the pickup marker. Only one cargo contract is active.
2. Accept	Enter a 22-unit horizontal radius; acceptance is automatic.
3. Load	Hold within pickup range for 1 second. Progress decays when drifting out.
4. Carry	Cargo attaches visibly; follow destination bearing, distance and beacon.
5. Unload	Enter a 20-unit horizontal radius and hold 0.75 seconds. Leaving resets unloading.
6. Settle	Pay the base reward once; add eligible bonuses to unsecured earnings; remove cargo. New contracts follow after a short linger and 14-second cooldown.

Failure and recovery rules

Traveling more than 520 units from an uncollected pickup abandons that contract. An accepted pickup times out after 60 seconds. Carrying for more than 240 seconds fails the delivery and frees the contract slot. Death fails cargo; extracting explicitly abandons an active delivery. Missing the shorter time-bonus deadline only loses that bonus.

Handling and services

Base cargo handling multiplier is 0.91, improved by Airframe ranks toward a 0.98 cap. In the active flight path it changes acceleration and climb demand. Pickup and unload tests use horizontal distance; they do not demand an actual landing or low altitude. Cargo health exists as a data field but a complete cargo-damage loop was not established.

Near either depot of the active contract, within 18 horizontal units, service restores 18% maximum hull, 20% maximum fuel and 25% of the selected weapon magazine when needed. The service cooldown is 16 seconds. The current function has no service price or landing requirement.

DESIGN NOTE

Godot decision: preserve arcade fly-over loading or require a clear altitude/speed envelope. If landing is required, teach it explicitly and show why loading is blocked.

Evidence: `delivery.ts`; `engine.ts`: `delivery` callbacks, `cargo` movement and `updateDepotService`

GAME DESIGN

Cargo values & reward examples

Cargo	Base value	Risk factor
Medical Supplies	150	1.05
Ammunition	170	1.12
Aviation Fuel	160	1.16
Electronics	215	1.08
Machine Parts	185	1.04
Military Supplies	245	1.20

How the contract price is calculated

Multiply the cargo base value by its risk factor, by **1 + min(1.15, distance / 900)**, and by contract difficulty: Standard 1.0, Risky 1.3 or High Value 1.65. Round to the nearest 10 credits. The speed bonus is approximately 25% of this reward, rounded to the nearest 5.

The time-bonus limit is the greater of 42 seconds or rounded(distance / 42 + 24). Standard contracts are used in waves 1-2. Risky routes become possible from wave 3; High Value from wave 5, with Electronics or Military Supplies selected for High Value cargo.

Worked example: 600-unit standard ammunition route

$170 \times 1.12 \times (1 + 600 / 900) = 317.33$, rounded to **320 secured credits**. The time bonus is **80** and the time limit is **42 seconds**. At High threat (1.25 reward factor), another **80** is a threat bonus. Finishing on time therefore produces 320 immediately banked credits plus 160 unsecured bonus credits, before any SAM exposure bonus.

SAM route risk

A carried contract can receive a one-time SAM exposure bonus, default 80 credits. It is an added route reward rather than a repeated payment for remaining near the launcher. In the engine settlement, time bonus, SAM risk bonus and threat bonus remain unsecured until extraction. Completed ammunition cargo also replenishes one flare.

DESIGN NOTE

The delivery announcement lists all reward components, but only the base price is immediately secured. The rebuilt HUD should label BANKED and AT RISK separately so players do not confuse a successful delivery with a fully secured run payout.

Evidence: delivery.ts: createDeliveryContract and markSamExposure; engine.ts: settleRewards

GAME DESIGN

Pickups, loot & resource recovery

Pickup	Current effect
Field Repair	Restore 35 hull before repair-efficiency effects.
Ammo Pack	Refill all weapon magazines.
Fuel	Restore 35 fuel, capped at tank capacity.
Shield	8-second protection timer.
Damage Overdrive	Double factory weapon damage for 8 seconds.
Speed Boost	6-second timer; speed run upgrade uses 12 seconds. Current flight integration needs correction.
Bomb / Airstrike	Destroy all active enemies and add 150 score; includes boss in current path.
XP Gem	Grant its stored run-XP value on collection.
Salvage	Add its quantity to salvage carried this run.
Countermeasure	Replenish flare capacity, capped at maximum.
Salvage Cache	Add 8 salvage and 250 score in the collection effect.
Magnet Surge	8-second wide pickup attraction.
EMP Pulse	Apply EMP within 65 horizontal units; divert active homing threats.

Drop design

Basic enemies use sparse bonus loot: 8% chance of salvage and 6% chance of a useful powerup in the loot planner. Special enemies and tanks have richer rolls. Elites drop 2-3 salvage and can add caches, useful items or flares. Boss loot guarantees 6 salvage, a cache, a useful powerup and a flare, in addition to direct boss rewards and XP.

XP values: basic 1, shooter 2, drone 3, tank 5, elite 15 and boss 50. Special variants use a threat-based 3-8 XP range unless elite/boss priority overrides it. Pickups animate into the helicopter, and nearby reward text is grouped to reduce clutter.

DESIGN NOTE

The contested salvage event creates a cache with a value of 10, but the cache effect awards a fixed 8. Decide whether caches use their stored value or a universal size, then make the label, payout and tests agree.

Evidence: loot.ts; logic.ts: xpForEnemyType; engine.ts: loot drops and applyPowerUp

GAME DESIGN

Run growth & upgrade choices

Run XP is separate from weapon XP and permanent credits. XP gems and mission rewards raise the run level from 1 to a maximum of 15. Cumulative XP thresholds start at 10, 25, 45, 70 and 100; level 15 requires 595. A large reward can queue multiple level-ups without losing choices.

Each level-up pauses gameplay and offers three distinct options, preferring different categories when possible. The chosen effect applies to this run. The following catalog includes active effects and legacy options that need compatibility work.

Upgrade	Effect / current design
Overclock Rounds	+25% all-weapon damage per stack.
Hair Trigger	+18% firing-rate factor per stack.
Extended Mag	Multiply magazine capacity by 1.3 and refill.
Speed Loader	25% faster reload wording; applied through reload bonus logic.
Salvo Overclock	Cooldown -35% per stack; minimum 1.5 seconds.
Reinforced Hull	+20 maximum hull and repair 20.
Reactive Armor	+6% mitigation per stack within the shared 24% cap.
Field Repair	Repair 25 and improve future repair efficiency.
Turbine Tune	Reduce fuel drain by 30% per stack; run factor floors at 0.4.
Magnetic Winch	Increase XP pickup attraction range by 30% per stack.
Aegis Field	Immediately grant an 8-second shield.
Afterburners	Advertised +20% speed for 12 seconds; current speed-cap integration is incomplete.
Vector Jets	Reduce dash cooldown 15%; dash binding is retired.
Airstrike	Trigger Bomb effect immediately.
Incendiary / EMP / Shock	+6% burn / +7% blast-weapon EMP / +5% shock proc chance per stack.

DESIGN NOTE

Do not ship choices that have no effective player benefit. Restore a supported dash/speed design or replace those offers before balancing the Godot upgrade pool.

Evidence: logic.ts: run XP and UPGRADE_POOL; engine.ts: applyRunUpgrade and effect use sites

GAME DESIGN

Hangar: permanent aircraft upgrades

The hangar spends the shared credit wallet on eight systems, each with five ranks. Purchases validate affordability and cap the rank. Effects should be readable as a before/after preview; insufficient funds and maximum rank should be explicit.

System	Rank 1 → 5 prices	Purpose / implementation note
Engine	190 / 460 / 810 / 1240 / 1740	Advertised speed and afterburner efficiency. Active fuel drain benefits up to 12%; current flight speed effect needs integration.
Rotor	180 / 440 / 780 / 1180 / 1660	Advertised climb and vertical acceleration; verify contribution in the current controller.
Armor	210 / 510 / 880 / 1330 / 1850	+10 maximum hull/rank and contribution to capped mitigation.
Airframe	195 / 480 / 840 / 1280 / 1780	Cargo handling improvement; stability/collision promises need checking against the active movement path.
Fuel	200 / 490 / 860 / 1300 / 1800	+4 fuel capacity and 6% lower base consumption per rank.
Targeting	195 / 480 / 860 / 1320 / 1840	Auto-aim range increases by 10 units/rank; tracking promises require alignment.
Weapon System	220 / 540 / 930 / 1400 / 1930	Magazine +4.5%/rank; reload duration -2.8%/rank, to 0.86x.
Countermeasures	170 / 430 / 780 / 1230 / 1830	Flare charge, cooldown, window and effectiveness milestones.

Aircraft identity

The menu offers Apache, Nighthawk and Warlock. Preserve the visual options, but treat balanced/agile/heavy performance differences as intended design until implemented in the active flight controller. Choose meaningful tradeoffs, such as acceleration versus hull capacity, only as an explicit rebuild decision.

Permanent and temporary layers

Hangar ranks and weapon-mod selections persist. Run upgrades, current resources, active weapon levels and temporary status timers reset. Mastery records the highest historical weapon level. Keep these layers distinct in the interface and in save behavior.

Evidence: delivery.ts: HANGAR_UPGRADE_INFO; engine.ts: resetWeaponsFromHangar; App.tsx; entities.ts

GAME DESIGN

Pilot perks & long-term motivation

Eight pilot perks have five ranks each. They use the same credit wallet as the hangar. Although some source descriptions say salvage credits, there is no separate persistent salvage-currency wallet: carried salvage converts to ordinary credits on extraction.

Perk	Effect per rank	Five rank prices
Magnet Radius	+15% pickup/gem attraction radius	100 / 220 / 420 / 700 / 1100
Vector Thrusters	-8% dash cooldown; input support unresolved	110 / 240 / 450 / 750 / 1150
Scavenger Rig	+10% salvage gain	120 / 260 / 480 / 800 / 1200
Rotor Armor	+4% damage reduction contribution; total cap still applies	130 / 280 / 500 / 820 / 1250
Combat Computer	+8% run XP	130 / 280 / 500 / 820 / 1250
Lean Burn	-6% fuel drain	100 / 220 / 420 / 700 / 1100
Devastator Core	+15% super charge gain	140 / 300 / 520 / 850 / 1300
EMP Rounds	+4% eligible status-roll chance; also affects burn/shock despite the EMP label	140 / 300 / 520 / 850 / 1300

Devastation super

The meter holds 100 charge. A normal kill starts at 4 charge, an elite at 12, and a boss grants 40; ordinary/elite kills add up to 6 more from combo count. Devastator Core increases this gain. Activating the full meter grants 5 seconds of invulnerability and a 2.2x firing-rate factor, with a radial 240-unit damage pulse. Charging is locked during the activation and subsequent 8-second cooldown.

First-time achievements

One-time payouts per profile: first kill 20 credits; wave 1 cleared 40; first delivery 60; first SAM destroyed 50; first extraction 80. These are settled with the run report and remembered to prevent repeated restart farming. High score, ten recent runs and weapon mastery supply additional personal goals.

DESIGN NOTE

The super has a working touch activation path but no current desktop key. Before selling super-related perks in the rebuilt game, provide a visible and usable action on every supported input device.

Evidence: logic.ts: PERK_INFO, achievements and super rules; engine.ts: activateDevastation; App.tsx

GAME DESIGN

Score, threat & the credit economy

Three distinct reward categories

Score measures the run. **Banked credits** are already in the wallet and survive death. **Unsecured bonuses and carried salvage** are at risk until extraction. Keep these concepts visually separate; a high score does not automatically mean a large secured payout.

Score and combos

Qualifying weapon hits build the combo counter and refresh a 3-second timer. Multiplier = $1 + \min(\text{combo} \times 0.1, 5)$, capped at 6x. Enemy score also uses low-hull risk and Endless Overdrive factors. Below 40% hull, the risk bonus rises toward a ceiling of +50% Casual, +75% Normal or +100% Hard as health approaches zero. Quick kills within 1.4 seconds create Double/Triple/Quad Kill, Killing Spree and Rampage announcements.

Threat	Points threshold	Reward factor
Low	0	1.00
Elevated	30	1.10
High	80	1.25
Critical	150	1.45
Extreme	240	1.70

Threat grows with time, kills, destroyed defenses and completed deliveries. It also raises composition pressure and spawn intensity. A threat bonus is $\text{base reward} \times (\text{factor} - 1)$, rounded; it is not a second full copy of the base reward.

Guaranteed combat pay

End-of-run pay = $2 \times \text{kills} + 12 \times \text{cleared waves} + 15 \times \text{counted objectives} + 4 \times \text{each completed 30 seconds}$ if at least one kill occurred. The current call counts SAM and radar destructions for this pay, excluding ammo depots. Example: 20 kills, wave 4 reached, 2 counted objectives and 120 seconds gives $40 + 36 + 30 + 16 = 122$ credits.

DESIGN NOTE

A helper describes one credit per 100 score, but it is not the active run-end wallet payout path. Use combat pay, mission rewards, deliveries, bounties and extraction settlement as the economy baseline.

Evidence: logic.ts: combo, risk, combat pay; mechanics.ts: threat; engine.ts: dispatchGameOver

GAME DESIGN

Extraction, failure & saved progress

The normal extraction opportunity

The run qualifies at wave 6 or later with at least two destroyed installations. Only one normal LZ offer is made per run. A marked landing zone prefers a roof or suitable ground surface. The player must remain within the zone and within 48 vertical units of it while extraction progress fills.

Normal hold time is 12 seconds; Casual progresses 1.18x and Hard 0.94x, giving approximately 10.2 and 12.8 seconds. Entering the zone adds combat pressure and reinforcement demand. Leaving reduces progress at 0.18 per second. An inactive offer can disappear after 90 seconds or when it is far behind the player. Active cargo is abandoned on successful extraction.

What is kept and lost

Extraction banks all unsecured credits and converts carried salvage at **5 credits per salvage**. Example: a 500-credit wallet, 160 unsecured credits and 8 salvage becomes 700 credits before additional end-of-run combat/achievement pay. On destruction, the 160 unsecured and 8 salvage are lost, while the 500 banked credits remain; eligible combat and achievement pay is still awarded.

Saved profile

The project remembers wallet credits, eight hangar ranks, eight perk ranks, aircraft selection, weapon mods, mastery records, high score, settings, tutorial completion, one-time achievements and up to ten run-history records. Current run state is restarted rather than resumed as a saved in-progress campaign.

Results and navigation

The debrief shows outcome, score, wave, survival time, kills, accuracy, combo, completed activities, secured/lost earnings and cause of death. Players can retry, return to the menu or hangar, inspect recent runs and copy a text scorecard. Pause provides resume, restart, settings and return-to-menu actions.

DESIGN NOTE

A tall rooftop LZ may conflict with the absolute 60-unit flight ceiling. In Godot, only offer reachable pads and test every generated LZ against the flight envelope. The existing 48-unit vertical tolerance should be made explicit or replaced with a clearer landing rule.

Evidence: logic.ts: extraction gates and history; engine.ts: extraction and run-end; App.tsx

GAME DESIGN

World design & environmental interaction

A continuous procedural battlefield

The city is assembled around the player using a repeating district schedule. The current list contains nine distinct districts, despite older comments mentioning ten. District identity changes building density, height, palette, roads, landmarks, rooftops and open-space dressing. Locations remain stable for a given region identifier.

District	Identity and navigational role
Downtown / Midtown	Tall dense blocks, towers and urban streets; cover and constrained flight lanes.
Industrial	Warehouses, factories, cooling towers and cargo facilities.
Residential	Apartment blocks, lower urban massing and clock-tower identity.
Waterfront	Marina, docks, water edges and strong depot availability.
Military Base	Radar installations, military clutter, barricades and landing infrastructure.
Desert	Sparse structures, open sand and mesa landmark.
Forest	Vegetation, sparse buildings and giant-tree landmark.
Ruins	Broken urban forms, industrial debris and fallen-tower landmark.

Rhythm and destruction

Dense streets alternate with open spaces, objective clearings and rarer landmarks. Building shapes include stepped towers, slabs, offices, warehouses, factories, residential blocks, parking and communications structures. Damage triggers staged collapse, debris and smoke; destroyed buildings no longer block routes. This is selective gameplay destruction, not a promise that every decorative object is fully destructible.

Living detail and tactical hazards

Traffic cruises, brakes, swerves and honks when buzzed. Animated signs, birds, rooftop fixtures, beacons, street furniture, palms, bridges and canals create scale. Volatile tanks/generators can chain-react: explosions harm nearby enemies and can also damage the player. Rooftop turrets create incidental combat pressure.

Weather and night

Storms darken the sky, change fog, add rain and lightning. Later waves increase storm intensity; night operations shift lighting and visibility. Wind is calculated, but its old flight-drift path is disabled. Preserve atmospheric effects and explicitly decide whether wind should affect the new controller.

Evidence: city.ts; props.ts; materials.ts; particles.ts; logic.ts: districts; entities.ts

GAME DESIGN

Interface, feedback & accessibility

Information hierarchy

Immediate survival: hull, fuel, current ammunition, reload, flares, missile warnings and crosshair. **Current task:** mission title/progress/bonus, cargo action/distance/reward and extraction progress. **Run progress:** wave, XP, level, score, combo, threat, wallet, unsecured credits, salvage and super charge.

The tactical minimap marks enemies by role, elites, priority targets, boss, installations, SAM detection zones, incoming homing missiles, delivery endpoints, missions and extraction. Off-screen pointers and distance/bearing labels guide the player toward tasks. Rooftop extraction indicators also convey height difference.

Impact and combat readability

Damage numbers group rapid hits and shotgun pellets. Reward text combines nearby XP/salvage events; elite and boss kills use larger grouped callouts. Hit markers, directional damage feedback, near-miss cues, lock markers, recoil, tracers, smoke, sparks, debris, shockwaves and short hit-stop communicate cause and consequence. Low flight creates rotor dust and a moving dust wake.

Audio functions

The sound palette covers rotor load, gun/shotgun fire, rocket/missile launches, reloads, explosions, damage, pickups, SAM locking, flare release, lock break, near misses, enemy fly-bys, tank cannon, radar collapse, electrical disruption, combos, upgrades, purchases, errors, warnings, wave transitions and boss introductions. Music and effects have separate levels under master volume.

Menus and settings

Menu functions include launch, aircraft selection, hangar tabs, how-to-play, settings and results/history. Settings cover difficulty, aim assistance, touch mode, controller sensitivity/inversion, handling selection, visual quality, retro SP1/HD appearance, adaptive quality, screen shake and reduced flashes. Tutorial can be replayed; progress reset requires an explicit confirmation.

DESIGN NOTE

Rebuild acceptance: warnings must remain understandable with sound muted, reduced flashes enabled and shake disabled. Avoid relying only on color. Keep the central combat view clear and ensure controller/touch users can operate every menu and essential combat action.

Evidence: App.tsx; index.css; audio.ts; floatingText.ts; engine.ts; particles.ts

GAME DESIGN

Design gaps to resolve before rebuilding

These are findings from source-path comparison, not claims of a complete manual playthrough. Existing automated checks pass, but they do not prove that every displayed feature is integrated into the live game.

Priority	Mismatch or uncertainty	Recommended Godot decision
High	Dash controls retired; dash perks and run upgrades remain.	Restore a documented action or replace every dependent offer.
High	Speed boost/model handling/weather drift remain in older movement paths.	Apply chosen modifiers to one active flight calculation; remove unsupported promises.
High	Super activation exists on touch but lacks a desktop binding.	Add remappable desktop/controller activation and consistent prompts.
High	Touch/controller combat and altitude actions lack full parity.	Complete an input-action checklist before platform release.
High	Bomb instantly removes bosses.	Choose explicit boss damage policy and test it.
High	Rooftop LZ placement versus 60-unit ceiling.	Validate reachability before offering extraction.
Medium	Mastery kill credit goes to the last-fired weapon.	Attribute rewards to the actual damage source.
Medium	Radar blast lethal-hit rewards need verification.	Route all deaths through a once-only settlement function.
Medium	Mission names suggest convoy defense/rescue simulation beyond proximity timers.	Keep arcade rules and label them clearly, or explicitly add health/crew systems.
Medium	Weapon XP comments show 4/10/18/28; actual formula gives 6/14/24/36.	Use one balance table for both rules and interface.
Medium	Help altitude range, rocket guidance and cache quantities disagree with behavior.	Use current flight envelope, explicit guidance and consistent item values.
Medium	Comments imply distinct aircraft handling and full rigid-body destruction.	Define true aircraft tradeoffs and selective destruction scope.

Additional cleanup: distinguish temporary Damage Overdrive from Endless Overdrive; ensure expired linked cargo contracts cannot leave a tactical mission stuck; audit pause behavior for all timers; make claimed reward amounts match banked versus unsecured earnings.

Evidence: Cross-check of current engine paths, data, help text and project documentation

GAME DESIGN

Functional specification: action contracts

This catalog describes game functions in plain language. Each contract identifies its input or trigger, the state change, and the feedback the player needs. It is the functional checklist for a rebuild, rather than a list of programming methods.

Function	Expected contract
Start / restart run	Load persistent choices; reset temporary state, threats, waves, entities and timers; show tutorial/opening and enter play.
Move / aim	Read chosen input source; apply heading, thrust, strafe and lift; collide with terrain; aim turret independently; follow with camera.
Fire / reload / switch	Check selection, cooldown, ammo and reload; launch valid shots; update ammunition; play the correct cue.
Paint / launch salvo	Acquire bounded locks while held; discard dead locks; release missiles; start cooldown and clear markers.
Deploy flares	Validate charge/cooldown; consume one charge; create decoy; retarget eligible missiles; show result.
Take damage / repair	Respect protection and mitigation; change hull and damage presentation; record source; trigger death or emergency repair once.
Destroy target	Resolve score, XP, bounties, mission progress, threat, drops, death effects and attack-slot release once.
Collect pickup	Move item into collection state; grant its effect/value once; update meters and grouped feedback; remove item.
Level up / choose	Queue every level crossed; pause combat; offer three distinct upgrades; apply one; resume or show next queued choice.
Advance wave / spawn	Update time and theme; refresh budgets; choose valid enemy groups; obey gates, caps, telegraphs and lulls.
Mission / cargo interaction	Track the correct target or location; advance hold/kill progress; evaluate bonus/failure; settle once; release the activity slot.
Extract / die / debrief	End combat once; bank or lose at-risk rewards; grant eligible combat pay and achievements; save records; show next actions.

Supporting functions: purchase upgrades/perks; choose aircraft/mods; change and persist settings; pause/resume; save/load safe defaults; reset progress; replay tutorial; copy scorecard; update minimap and warnings; generate world regions; collapse buildings; animate traffic, weather, night and effects.

Evidence: Player-facing synthesis of gameplay and interface functions

GODOT DEVELOPMENT

Godot roadmap: sequence & scope

Target: Godot 4.7.2 stable. The official Windows download page identifies this release, dated 18 August 2026, as the current download on the review date. Preview builds are separate. Recheck the official stable version when development starts, and keep editor and export templates matched. [G1]

The phases below are proposed work packages. They translate the reviewed design into a playable Godot game. They do not describe work already completed. No calendar estimate is assumed: team size, art scope, target hardware and platform commitments are not supplied.

Phase	Deliverable	Depends on
1	Approved rules, baseline scenes and input plan	This design review
2	Player flight, independent turret and camera	1
3	Weapons, damage and a first combat slice	2
4	Enemy roles and coordinated encounter director	3
5	World districts, destruction and atmosphere	2-4
6	SAM/radar tactics and countermeasures	3-5
7	Missions, cargo and depot services	5-6
8	XP, loot, hangar economy and saved profile	3-7
9	Boss, extraction and Endless Overdrive	4, 6-8
10	Complete interface, tutorial and accessibility	2-9
11	Balance, integration and performance validation	1-10
12	Release candidate, exports and final acceptance	11

Milestones worth playing

First combat slice: one helicopter, one district, two enemies, four weapons, damage, death and restart. **Complete run:** waves, a delivery, a tactical mission, rewards and extraction. **Content complete:** roster, districts, boss, mods, perks and all settings. **Release candidate:** input parity, persistence, readability and long-session stability verified.

DESIGN NOTE

Use the current rules as the baseline. Any new landing system, aircraft tradeoff, boss resistance or deeper convoy simulation is a proposed design change and should be recorded separately from faithful reproduction.

Evidence: Proposed development sequence | Godot official download [G1]

GODOT DEVELOPMENT

Phase 1 | Freeze the rules & start the project

Goal: Create a clear, testable baseline before producing a large amount of content.

Build in this order

- Create the Godot 4.7.2 project and a small flight test level. Establish separate playable, menu and results scenes so a run can start and end cleanly.
- Create a decision register for the high-priority gaps: dash, super bindings, speed effects, aircraft differences, boss Bomb response and reachable extraction. Mark each as preserve, repair or redesign.
- Create editable balance records for weapons, enemies, missions, pickups and progression. Use the values in this PDF as the first baseline; keep explanatory names beside numbers.
- In Project Settings → Input Map, create named actions for yaw, thrust, strafe, climb, descend, aim, fire, salvo, flares, weapon selection, reload, super and pause. Assign keyboard, controller and touch equivalents. [G2]
- Define run states: menu, tutorial, countdown, active, paused, upgrade choice, boss decision and results. Decide which clocks progress in each state.

Completion checks

- A fresh profile can enter and leave a test run without carrying old input or temporary state.
- Every essential action has an agreed mapping, and no help text advertises an unimplemented action.

Reviewable deliverable

A tiny launchable project, an action map and an approved list of gameplay decisions. No final art is required.

DESIGN NOTE

Dependency / next step: All later phases use this rule baseline. Avoid balancing an effect until it is connected to active gameplay.

Evidence: Proposed plan; Godot InputEvent/Input Map documentation [G2]

GODOT DEVELOPMENT

Phase 2 | Flight, aiming & camera

Goal: Make flying and aiming satisfying before adding combat complexity.

Build in this order

- Build a reusable helicopter scene with collision shape, visual hull, rotor, independent yaw/pitch gun pivots and muzzle marker. Keep visual bank separate from the heading used by movement.
- Implement forward/reverse thrust, yaw, strafe, drag and collective lift in a controlled 3D character-motion setup. Start with 72-unit cruise, 1.85-radian yaw, 6.5/7 climb/sink and a 60-unit ceiling.
- Add surface clearance and solid-building avoidance. Place low roofs, narrow streets and towers above the ceiling in the test level. Verify diagonal movement does not increase top speed.
- Aim the gun at a world-space mouse target or controller/touch aim direction. Make projectiles later originate from the real muzzle. Separate aircraft steering from turret rotation.
- Add the tail camera, speed-dependent pull-back, look-ahead, restrained bank and occlusion handling. Connect any approved speed, cargo, aircraft or handling modifiers to the same active movement calculation.

Completion checks

- Hover, reverse, strafe, turn 180 degrees and release controls at low and high frame rates; the result remains stable.
- Fly toward buildings and above roofs; no tunneling, camera disappearance or elevation above the ceiling is allowed.
- Aiming while turning does not snap the helicopter toward the cursor; each chosen aircraft difference can be measured.

Reviewable deliverable

A flight playground with a repeatable route and a written feel comparison against the supplied game.

DESIGN NOTE

Dependency / next step: Requires Phase 1. Weapon and enemy development should use this stable flight envelope.

Evidence: Proposed implementation and acceptance criteria

GODOT DEVELOPMENT

Phase 3 | Weapons & first playable combat

Goal: Complete a small but fully playable fight with correct hit and reward ownership.

Build in this order

- Build the four weapons from the factory table. Add magazine counts, trigger interval, switching, manual/automatic reload and distinct muzzle/launch feedback.
- Create direct shots, homing missiles, assisted or ballistic rockets according to the approved decision, and multi-pellet shotgun fire. Use collision sweeps suitable for fast projectiles so shots cannot skip thin targets.
- Create one infantry target and one aerial target. Apply hull, target categories, damage affinity, shields, repair and player damage through shared rules.
- Tag damage with its owner, weapon and cause. Resolve destruction once; this must also work for delayed missiles, burn damage, chain explosions and Bomb effects.
- Implement the six-lock salvo, three-lock per-target cap, lock cadence and cooldown. Add death/restart and a compact hull/ammo HUD to complete the combat loop.

Completion checks

- Compare damage, firing interval, ammo spent and reload time against the baseline; six shotgun pellets still spend one magazine unit.
- A dead target cannot pay twice when hit by multiple projectiles on the same update.
- Firing into a wall, changing weapon while a missile is in flight, and releasing salvo after a target dies all behave safely.

Reviewable deliverable

A short combat slice: launch, maneuver, aim, fight, reload, take damage, die and restart.

DESIGN NOTE

Dependency / next step: Requires Phase 2. Do not add the full enemy roster until this slice is fun and reward attribution is reliable.

Evidence: Proposed implementation and acceptance criteria

GODOT DEVELOPMENT

Phase 4 | Enemies & encounter direction

Goal: Build pressure through varied decisions while keeping incoming attacks readable.

Build in this order

- Implement infantry, tank and light aircraft first. Ground units use terrain placement and lanes; tank turret aim is independent of its chassis; aircraft use the full attack-run sequence.
- Add attack permissions for air, ground and heavy threats. Return permissions on death, cancellation and timeout. Add directional sectors, attacker cooldowns and short collection lulls.
- Introduce variants in families: fast attackers, missile/rocket threats, shield/repair support, heavy hulls, interceptors and minelayers. Use distinct silhouettes and warning cues for each.
- Apply elite, shielded, regenerating, explosive, splitter and vampiric effects. Give every status a clear indicator and prevent stacked effects from creating unreadable or unavoidable attacks.
- Add time-driven waves, population targets, budgets, difficulty multipliers, wave themes and event queues. Keep normal gates separate from explicit event exceptions.

Completion checks

- One aircraft completes approach, setup, attack, break-away and re-entry against a stationary or moving player.
- Three and eight aircraft respect simultaneous attack limits; a tank/SAM combination cannot overbook heavy attacks.
- Shield and repair units change the fight visibly; killing them removes their contribution immediately.

Reviewable deliverable

A ten-wave encounter test course with selectable enemy groups, readable tells and measured active-attacker counts.

DESIGN NOTE

Dependency / next step: Requires Phase 3. Use placeholder art until the behavior is accepted; world layout and defenses follow.

Evidence: Proposed implementation and acceptance criteria

GODOT DEVELOPMENT

Phase 5 | Districts, cover & destruction

Goal: Turn the test arena into the game's navigable, changing battlefield.

Build in this order

- Build modular district regions with streets, clear flight corridors, building footprints, rooftops and reserved objective/depot locations. Start with Downtown and Industrial, then expand to all nine districts.
- Use stable region seeds so a location, depot and landmark remain predictable when the player leaves and returns. Blend seams between regions and test both travel directions.
- Add building damage states and staged collapse. Update collision and usable surface height when a building is destroyed; create temporary debris and remove it after its visual job is finished.
- Add volatile fuel/generator props with bounded chain reactions and shared damage attribution. Add traffic, signs, birds, palms, road dressing and landmarks after routes are playable.
- Add clear daylight, storm transitions, rain, lightning and night palettes. Connect weather to flight only if that design was approved; keep visibility sufficient for attack warnings.

Completion checks

- Long-distance flight and a return trip retain the same route endpoints and do not create visible gaps or unexpected walls.
- Destroying cover opens the intended route, removes attached hazards and never repeatedly pays rewards.
- The helicopter and imminent threats remain readable in dense streets, storms and night scenes.

Reviewable deliverable

A repeatable multi-district route with destruction, a depot corridor and atmosphere previews.

DESIGN NOTE

Dependency / next step: Requires stable movement and combat. Objective, cargo and extraction placement depend on validated world surfaces.

Evidence: Proposed implementation and acceptance criteria

GODOT DEVELOPMENT

Phase 6 | SAMs, radar & defensive play

Goal: Create understandable area-denial threats and reliable counterplay.

Build in this order

- Build the SAM installation with search, track, lock, fire, reload, disabled and destroyed states. Show alignment and increasing lock urgency before launch.
- Implement homing missiles with lifetime, turn limits and a visible threat indicator. Show direction and danger; distinguish missiles targeting the player from those diverted to a decoy.
- Implement flares with charges, cooldown, active window and distance-based diversion. Add replenishment and rank-based improvements without silently making distant launches harmless.
- Add radar support to nearby SAMs and implement destruction effects for SAM, radar and ammo depot. Ensure explosion kills feed the same scoring and loot settlement as weapon kills.
- Place mixed encounters behind cover and along cargo routes. Add rooftop turrets separately from the major installations and remove them when their building collapses.

Completion checks

- Leaving the SAM envelope or obstructing required alignment cancels a lock; destroyed SAMs cannot launch later.
- A close eligible missile is diverted, a distant missile can remain dangerous, and cooldown/empty-charge presses do not spend extra charges.
- Radar destruction removes its support effect and credits lethal damage exactly once.

Reviewable deliverable

A defense encounter that can be won by destroying support, using cover, dodging or timing flares.

DESIGN NOTE

Dependency / next step: Requires Phases 3-5. Integrate these hazards into missions and cargo routes next.

Evidence: Proposed implementation and acceptance criteria

GODOT DEVELOPMENT

Phase 7 | Missions, deliveries & services

Goal: Add optional activities that create route choices during survival combat.

Build in this order

- Create a reusable mission record with target identity, completion progress, optional bonus, rewards and state. Add SAM, radar, elite hunt and area-clear missions first.
- Add defend, rescue and escort proximity objectives. Represent the convoy/crash site clearly. If actual convoy health or crew transport is added, record that as new scope with its own failure rules.
- Implement a separate cargo contract slot and the offer/accept/load/carry/unload/complete flow. Preserve deterministic endpoints, distance limits, time bonus and failure recovery.
- Attach a visible cargo pod and apply its handling effect. Show pickup/delivery direction, remaining distance and loading progress. Add depot repair/fuel/ammo service with cooldown.
- Implement main and optional rewards independently. When a linked cargo or target disappears, resolve the mission explicitly so it cannot block the next assignment.

Completion checks

- Complete every mission and deliberately fail every optional bonus; bonus failure must not cancel a valid main reward.
- Overshoot a pickup, leave an unload zone, miss a time bonus, carry too long, die and extract with cargo; every case frees or resolves the proper state.
- Repeat completion events and save reloads; no reward is granted twice.

Reviewable deliverable

A combat session with one tactical mission and one cargo contract active together, including clear reward breakdowns.

DESIGN NOTE

Dependency / next step: Requires world placement and defense systems. Its reward events become inputs to progression in Phase 8.

Evidence: Proposed implementation and acceptance criteria

GODOT DEVELOPMENT

Phase 8 | Progression, loot & saving

Goal: Make every completed activity contribute to understandable short- and long-term growth.

Build in this order

- Implement XP gems, pickup attraction, all resource items, salvage caches and countermeasure drops. Attribute loot to one settled death and delay crash loot only when the death animation requires it.
- Implement run levels 1-15 and three-choice upgrade cards. Queue multiple levels from one reward. Remove or repair inactive dash/speed options before they enter the offer pool.
- Implement weapon levels 1-5, correct cumulative XP thresholds, signature effects and per-weapon mods. Store historical mastery separately from current-run levels.
- Build hangar purchases and pilot perks with rank caps, prices, affordability feedback and before/after values. Wire each purchased effect into active gameplay and demonstrate its benefit.
- Save persistent choices and rewards in a versioned profile under Godot's user-data location. Validate loaded data, recover safe defaults and preserve a last valid save where practical. Save only the persistent profile unless a new in-progress save feature is explicitly approved. [G4]

Completion checks

- A fresh profile receives the starter grant once; achievements pay once; purchases survive relaunch.
- A large XP reward offers every earned choice, and pause does not drain fuel or expire statuses.
- Missing, malformed or older save data cannot create negative money, impossible ranks or a blocked menu.

Reviewable deliverable

A playable earn-spend-relaunch loop with a transparent wallet and all permanent effects demonstrated.

DESIGN NOTE

Dependency / next step: Requires combat and mission reward events. Extraction settlement must use the same wallet rules.

Evidence: Proposed plan; Godot saving-games guidance [G4]

GODOT DEVELOPMENT

Phase 9 | Boss, extraction & endless mode

Goal: Give the run clear endpoints and a meaningful continue-or-leave decision.

Build in this order

- Build the heavy gunship with the three hull stages, engine damage cues, telegraphed attacks and escorts. Reduce background combat pressure during the encounter.
- Implement the approved Bomb-versus-boss rule. Ensure boss defeat, explosion effects, score, drops and repair occur once even if multiple sources hit together.
- Offer a reachable normal LZ after wave/objective eligibility. Validate position, height, approach space and the exact hold zone; show extraction progress and incoming pressure.
- On successful extraction, bank unsecured bonuses and salvage conversion; on death, remove them. In both cases settle eligible combat pay and achievements exactly once.
- After boss defeat, pause for a clear safe-exit or Endless Overdrive choice. Continue from wave 11, scale pressure within caps and define the behavior of each later boss decision.

Completion checks

- Cross 66% and 33% boss hull boundaries and verify the correct attacks and telegraphs.
- Extract with cargo, leave and re-enter the hold zone, let an offer expire and attempt every rooftop pad; no unreachable or stuck state is acceptable.
- Use known wallet/bonus/salvage totals to verify death and extraction arithmetic, then relaunch to confirm saved balances.

Reviewable deliverable

A complete run ending in destruction, normal extraction, boss extraction or voluntary endless continuation.

DESIGN NOTE

Dependency / next step: Requires enemy direction, defenses, activities and progression. Next make every rule legible through the interface.

Evidence: Proposed implementation and acceptance criteria

GODOT DEVELOPMENT

Phase 10 | Interface, teaching & accessibility

Goal: Make the complete game understandable without reading this document.

Build in this order

- Build menu, hangar, armory/mods, perks, settings, help, pause, level-up choice, boss decision and results screens. Define controller focus order and touch-safe buttons.
- Build the combat HUD in survival/task/progress layers. Add the minimap, off-screen pointers, missile danger, mission bonuses and clearly separated banked versus at-risk earnings.
- Update tutorial prompts from the actual input map. Teach movement and aiming before the first serious threat, then flares, salvo, resource recovery and extraction. Allow skip and replay.
- Add distinct sound cues and restrained visual feedback. Provide independent volume controls, shake off/low/full, reduced flashes and readable contrast.
- Use Godot pause/process modes so combat stops while pause and upgrade menus remain interactive. Keep gameplay timers separate from menu animation and transition clocks. [G3]

Completion checks

- Complete a full run with keyboard/mouse, then controller, then touch if touch is a release target; all essential actions and menus must be accessible.
- Pause during a missile lock, reload, cargo hold, buff and extraction; resume without surprise expiry or duplicate input.
- Mute audio and disable shake/flashes; warnings and damage direction remain clear.

Reviewable deliverable

A first-time-player build with complete help, input parity, accessible settings and readable debrief.

DESIGN NOTE

Dependency / next step: Requires the full feature set. New-player observation now drives balance and polish.

Evidence: Proposed plan; Godot pause/process-mode documentation [G3]

GODOT DEVELOPMENT

Phase 11 | Balance & integration

Goal: Verify that the systems work together and every choice has a useful role.

Build in this order

- Run controlled scenarios with baseline and upgraded profiles on each difficulty. Measure time to kill, damage taken, fuel pressure, mission duration and credits earned per run.
- Compare weapon performance against light, armored, air and boss targets. Check stacking of mastery, mods, status procs and temporary damage; prevent one option from eliminating all alternatives.
- Observe new players through the tutorial and first three waves. Track misunderstood controls, unseen threats, abandoned cargo and purchases whose benefit is unclear.
- Stress late waves with support units, SAMs, projectiles, collapsing buildings, loot and weather together. Reduce decorative load before reducing threat cues or input response.
- Repeat long sessions and many restarts. Check that enemies, pickups, effects, timers and audio stop correctly and do not accumulate between runs.

Completion checks

- Early defeat still pays understandable progress; idling without contribution cannot farm time pay.
- No advertised upgrade is ineffective, and every enemy attack has readable counterplay at the supported performance floor.
- Rewards, mission state, boss state and extraction agree after unusual event ordering and repeated restarts.

Reviewable deliverable

A balance report with measured results, the remaining issues ranked by player impact, and a stable content-complete build.

DESIGN NOTE

Dependency / next step: Requires Phases 1-10. Freeze new gameplay scope before final release testing.

Evidence: Proposed implementation and acceptance criteria

GODOT DEVELOPMENT

Phase 12 | Release candidate & handoff

Goal: Deliver a reliable build with verified player flows and a maintainable design baseline.

Build in this order

- Lock the release rules and record final values for weapons, waves, rewards, inputs and purchased effects. Update the help and tutorial to those exact rules.
- Create exports for the approved target platforms using matching Godot export templates. Test the exported game on representative devices rather than only in the editor.
- Verify first launch, saved-profile migration, settings persistence, audio start, focus loss, pause/resume, controller disconnect/reconnect and clean exit.
- Run the acceptance matrix on the next page. Investigate any disagreement between displayed rewards and saved balances before calling the build complete.
- Package the playable build with a concise player guide, known limitations, final balance sheet and design decision log. Keep the Godot project and original project reference available for future revisions.

Completion checks

- A fresh player can learn, complete a mission/delivery, progress, extract or defeat the boss, spend earnings and relaunch successfully.
- Every supported platform passes the input and readability requirements; unsupported features are not advertised.
- No high-priority gameplay blocker, unreachable objective or duplicate reward issue remains.

Reviewable deliverable

A release candidate and a signed-off gameplay checklist. Release readiness depends on these outcomes, not simply on reaching the final phase.

DESIGN NOTE

Dependency / next step: Requires Phase 11. Add future content only after this baseline is stable.

Evidence: Proposed implementation and acceptance criteria

VERIFICATION

Acceptance matrix & test scenarios

Area	Scenario	Pass condition
Flight	Hover, strafe, reverse, climb, collide; vary frame rate.	Stable handling; capped altitude/speed; no wall passage.
Weapons	Fire every weapon; reload empty; switch during delayed damage.	Correct ammo/timing and source-owned damage/XP.
Defense	SAM with radar; close/distant flare; EMP; empty charges.	Readable lock sequence and correct diversion/cooldown.
AI	One, three and eight aircraft; ground + air + SAM.	Attack slots enforced; terrain avoided; valid counterplay.
Destruction	Collapse a turret building; chain volatile props.	Collision updates; hazards removed; deaths settle once.
Missions	All nine types; wrong target; failed optional bonus.	Only eligible progress counts; main and bonus separate.
Cargo	Overshoot, drift, timeout, die, extract while loaded.	No stuck cargo/mission; fair and correct settlement.
Growth	Large XP reward; all upgrades/mods/perks.	All choices delivered; every effect demonstrable.
Economy	Fresh grant, achievements, buy, die, extract, relaunch.	No duplication; exact banked/at-risk arithmetic.
Boss	Threshold crossings; Bomb; simultaneous lethal hits.	Approved stage rules; one reward and decision.
Interface	All input devices; mute; reduced flashes; small display.	All actions reachable; warnings remain understandable.
Stability	Long late-wave play; repeated restart/pause.	No accumulating entities, locked controls or timer drift.

Verification performed for this review

The existing project's automated suite completed successfully: **21 test files, 310 tests passed**. Coverage includes flight, camera, combat, AI, wave scaling, missions, deliveries, loot, feedback, world queries and model construction. Existing screenshots were inspected for visual context.

This document is based on source inspection, existing tests and project images. No complete manual end-to-end run across every input device or full ten-wave campaign was performed for this review. Passing existing tests does not resolve the design mismatches listed earlier.

Evidence: Existing automated suite executed during this review; proposed Godot acceptance criteria

REFERENCE

Source map & terminology

Local project evidence

Design area	Reviewed evidence
Game flow and interactions	src/game/engine.ts; src/App.tsx; src/game.ts
Flight, enemies and projectiles	src/game/entities.ts; src/game/enemyHelicopterModels.ts
Balance, progression and encounters	src/game/logic.ts; src/game/types.ts; src/game/combatDirector.ts
Damage, threat, defenses and loot	src/game/combat.ts; src/game/mechanics.ts; src/game/sam.ts; src/game/loot.ts
Activities and hangar	src/game/mission.ts; src/game/delivery.ts
World and presentation	src/game/city.ts; src/game/props.ts; src/game/materials.ts; src/game/particles.ts; src/game/debris.ts; src/game/floatingText.ts; src/audio.ts; src/index.css
Intent and validation	README.md; PROJECT_DETAILS.md; CHANGELOG.md; 21 gameplay test files; supplied screenshots

Current executable paths take precedence over older comments and documentation. Base values exclude modifiers unless explicitly stated. A defined feature can be partially connected, so the gap section identifies important integration limits. Source filenames are traceability references, not a technology-stack inventory.

Official Godot references - checked 28 September 2026

[G1] [Godot official Windows download and stable version](#)

[G2] [Input events and Input Map](#)

[G3] [Pausing games and process modes](#)

[G4] [Saving games and persistent data](#)

Useful terms

Hull: health. **Proc:** chance for a hit to trigger an extra effect. **Telegraph:** warning before an attack. **LZ:** landing/extraction zone. **SAM:** surface-to-air missile. **Salvage:** carried reward converted on extraction. **Banked:** already retained in the wallet. **At risk:** lost if the run ends in destruction.

All Godot phases, proposed fixes and acceptance criteria are recommendations derived from this project review. They are not claims that a Godot port already exists.

Evidence: Source index and official reference links